

KNOWLEDGE GRAPH AND INFORMATION RETRIEVAL SYSTEMS

Module	Content		PPTs	Book to refer to
1	Introduction to Information Retrieval and Graphs			
	1.1	Introduction to Information Retrieval: Concepts, data vs information, trends and research issues in information retrieval. The IR System architecture and retrieval process.	Part 1 Part 2 Part 3	T1, R1, R2, T6
	1.2	Query structures: keyword, pattern matching, structured queries. Structured knowledge Vs Unstructured knowledge		
	1.3	Introduction to Neural IR and Modern AI Retrieval: Dense retrieval, embedding-based search. Overview of multimodal embeddings (text, image, audio). Sparse vs Dense Retrieval: BM25 vs Dense Retrieval, scenarios where sparse retrieval performs better (legal, scientific and exact-match domains). Introduction to late interaction retrieval models such as ColBERT.		
	1.4	Advanced Attention Mechanisms for Retrieval AI: Grouped-Query Attention (GQA). Sliding Window Attention using Longformer. Flash Attention for scalable retrieval and long-context processing.		
2	Modeling in Information Retrieval, Indexing and Ranking			
	2.1	Review of IR models: Vector Space Model and Probabilistic Model. Transition from BM25 to dense vector representations. Neural ranking Retrieval architectures and semantic similarity search.	IR Models Problems on Probabilistic Model Problems on Vector Model	R1, R2, T4, R3
	2.2	Vector indexing techniques: Embedding generation, vector storage, Approximate Nearest Neighbor (ANN) algorithms (HNSW, IVF, Product Quantization). Hybrid Search techniques combining BM25 and Dense Retrieval. Metadata-aware search and filtering. Fine-Tuning and Alignment of Large Language Models.		
	2.3	Enterprise Retrieval Infrastructure: Elasticsearch, OpenSearch, PostgreSQL with pgvector, vector databases including Qdrant and LanceDB for enterprise-scale RAG systems.		
3	Cross-Modal Knowledge Graph Fundamentals			
	3.1	Knowledge Graph fundamentals: Representing knowledge as triples (subject–predicate–object). Nodes, edges, and properties. Introduction to multimodal knowledge graphs (text, image, audio nodes).		
	3.2	Graph models and Cross-Model design: Labelled Property Graph model. Cross-domain and cross-modal modeling.		

		Schema design, ontology basics, and avoiding modeling anti-patterns. Introduction to RDF (Resource Description Framework), OWL and SPARQL query language for semantic web and linked-data applications.		
	3.3	Model Evaluation and Explainability: Perplexity (Measuring model uncertainty in predictions) BLEU, ROUGE, METEOR (Text generation quality) BERTScore (Semantic similarity-based evaluation), LLM-as-Judge evaluation		
4	Multimodal Knowledge Graph Construction			
	4.1	Graph Databases and Architecture: Graph database concepts and architecture, Storage and indexing in graph databases,		
	4.2	Graph querying fundamentals, Traversal techniques and pattern matching in graphs. Knowledge extraction and relationship discovery. Entity Resolution and Deduplication techniques in knowledge graph construction. Schema-guided extraction using LLMs with structured output frameworks such as Instructor and Pydantic.		
	4.3	Document Understanding and AI-Based Extraction: OCR and LLM-based document understanding. Information extraction from multimodal documents.		
5	Advanced Retrieval Architectures			
	5.1	Advanced Chunking & Retrieval Optimization -Fixed-Length vs. Recursive Chunking, Semantic Chunking, IR Optimization		
	5.2	Advanced RAG Architectures: Contextual Retrieval (Anthropic), Corrective RAG (CRAG), Self-RAG, and Microsoft GraphRAG architecture and workflows, Multimodal RAG		
6	GraphRAG, Evaluation and Case Study			
	6.1	GraphRAG architecture with multimodal context. Query decomposition and reasoning paths using knowledge graphs. Agentic AI systems: tool usage (search engines, KG querying, APIs), multi-agent workflows.	Evaluation	
	6.2	Evaluation of retrieval systems: Standard test collections. Metrics: NDCG, Recall@K, Mean Reciprocal Rank (MRR), RAGAS framework. Assessing relevance and answer quality. LLM as a Judge.		
	6.3	Explainability for IR systems: Graph-based reasoning paths and interpretability. Enterprise knowledge graphs, geospatial analytics, multimedia information retrieval, distributed IR systems.		

Textbooks:	
1	Knowledge Graphs: Fundamentals, Techniques, and Applications Mayank Kejriwal, Craig A. Knoblock, Pedro Szekely Published by The MIT Press
2	Graph Database Engineering with Neo4j: Designing, Querying, and Scaling Connected Data Systems for Intelligent Applications, Dylan Yang, Packt Publishing / Neo4j Press, 2026
3	Essential GraphRAG , Tomaz Bratanic, Oskar Hane Published by Manning Publications, 2024
4	Vector Database Engineering, T. Larson, O'Reilly Media, 2025
5	Building Knowledge Graphs: A Practitioner's Guide Authors: Jesus Barrasa, Jim Webber Published by O'Reilly Media, Inc.
6	Hands-On Large Language Models , Jay Alamar, Maarten Grootendorst Publisher: O'Reilly Media, 2024
References:	
1	Introduction to Information Retrieval by Christopher D. Manning and Prabhakar Raghavan, Cambridge University Press.
2	Neural Information Retrieval , Bhaskar Mitra, Nick Craswell, ACM Synthesis Lectures, 2018
3	Designing Data-Intensive Applications , Martin Kleppmann, Published by O'Reilly Media
4	Multimodal Machine Learning, Tadas Baltrušaitis, Chaitanya Ahuja, Louis-Philippe Morency, Cambridge University Press, 2023
5	Building AI Agents with LLMs, RAG, and Knowledge Graphs , S. Raieli, G. Iuculano, Packt Publishing, 2025
6	Hands-On Large Language Models , Jay Alamar and Maarten Grootendorst, O'Reilly Media, 2024.

*SL: Self Learning Topic & Digital Resources (Content Beyond Syllabus) 60hrs	
1	Neo4j GraphAcademy (10–12 hrs) Learn graph databases, Cypher querying, graph modeling, and traversal techniques. Complete beginner and intermediate learning paths and submit badges/certificates.
2	LangChain Documentation & Tutorials (8–10 hrs) Build a basic Retrieval-Augmented Generation (RAG) pipeline using LangChain and vector databases. Evaluate using retrieval quality metrics and present as a mini case study.
3	LlamaIndex Documentation (6–8 hrs) Understand document indexing, chunking strategies, semantic retrieval, and GraphRAG pipelines. Implement a simple document-based QA system and submit report/demo.

4	Sentence Transformers Documentation (5–8 hrs) Explore embedding generation and semantic similarity using Sentence-BERT models. Compare dense retrieval with BM25 retrieval methods using sample datasets.
5	OpenAI Cookbook – RAG & Embeddings (5 hrs) Learn embeddings, hybrid retrieval, chunking, reranking, and RAG optimization techniques. Present findings through assignment or viva.
6	Hugging Face Transformers Course (10 hrs) Learn transformers, attention mechanisms, fine-tuning, zero-shot and few-shot learning. Earn course completion badge/certificate.
7	Vector Databases Explained – Pinecone Learning Center (5 hrs) Study vector indexing, ANN search, hybrid retrieval, and scalable vector database architectures. Prepare comparative study notes.
8	Advanced MOOCs and Online Learning Resources: Generative AI with Large Language Models – AWS & DeepLearning.AI Information Retrieval and Search Engines – NPTEL Multimodal AI – Carnegie Mellon University Resources
9	Knowledge Graphs for RAG (DeepLearning.AI)
10	Neo4j GraphAcademy – Knowledge Graph & GraphRAG
11	" GraphRAG : Unlocking LLM discovery on narrative private data" (Microsoft)
12	" Retrieval-Augmented Generation for Knowledge-Intensive Tasks"
13	Agentic AI Systems and Multi-Agent Frameworks (10–12 hrs) Study agent memory systems, MCP, ACP/A2A communication patterns, multi-agent orchestration, tool calling, adaptive reasoning, hooks/guardrails, and evaluation frameworks using LangGraph, CrewAI, AutoGen, RAGAS, and related open-source ecosystems. Implement a mini agentic workflow and prepare evaluation reports.

** SL includes Self learning, MOOCs, Spoken tutorials, online educational resources etc.
Total SL hours (60 hrs) to be asse*